Real-Time Action Recognition via 3D CNN and Optical Flow Fusion

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Abstract—*We present a real-time human action recognition system fusing spatiotemporal features from a lightweight 3D CNN (C3D-ResNet18) with explicit motion cues from a Farneback optical flow branch processed by a 2D ResNet-18. Trained and evaluated on UCF-101 under the standard three-fold protocol, the system achieves 87.2% Top-1 accuracy at 22.8 FPS on a single NVIDIA RTX 5060 Ti with FP16 inference and 8-frame clip subsampling, exceeding the real-time threshold of 15 FPS. Ablation experiments on Split 1 confirm that both branches contribute meaningfully and that the inference-time subsampling trade-off is modest (-1.2 points on Split 1). These results demonstrate that competitive accuracy and real-time throughput are simultaneously achievable without resort to large Transformer-based architectures.*

Index Terms—Action recognition, 3D CNN, optical flow, two-stream network, real-time inference, UCF-101.

I. Introduction

Human action recognition in video is a cornerstone problem in computer vision, with widespread applications in surveillance, human-computer interaction, sports analytics, and healthcare monitoring. Unlike static image classification, video understanding demands joint modeling of spatial appearance and temporal dynamics across consecutive frames. While significant progress has been made in offline systems, real-time action recognition remains a formidable challenge due to the high computational cost of processing dense video streams under latency constraints.

This paper develops and evaluates a real-time action recognition system on the UCF-101 benchmark. We propose a two-branch fusion architecture combining 3D CNNs for spatiotemporal feature extraction with Optical Flow-based CNNs for explicit motion representation. The primary research question is whether fusing these two complementary modalities can improve recognition accuracy while still meeting real-time inference requirements ($\geq$15 FPS).

II. Related Work

Early video recognition relied on hand-crafted descriptors such as HOG3D and improved Dense Trajectories (iDT). Simonyan and Zisserman [2] established the two-stream paradigm, demonstrating that explicit optical flow cues significantly boost accuracy. Tran et al. [1] proposed C3D, jointly learning spatiotemporal features via 3D convolutions. Feichtenhofer et al. [3] extended this with the SlowFast architecture, processing video at two temporal resolutions to capture slow semantic and fast motion features simultaneously.

More recently, Transformer-based architectures have achieved state-of-the-art results on large-scale benchmarks. ViViT [7] factorizes spatial and temporal attention over video tokens, while Video Swin Transformer [8] extends shifted-window attention to the temporal dimension, achieving strong results on Kinetics-400. However, their inference cost substantially exceeds lightweight CNN approaches, making real-time deployment on a single consumer GPU non-trivial. Our work targets this under-explored real-time regime, examining the speed-accuracy trade-off achievable with a fused 3D CNN and optical flow approach.

III. Proposed Method

A. Architecture Overview

We propose a dual-branch fusion network. The RGB branch ingests 16-frame clips at 224×224 resolution through a lightweight 3D ResNet-18 backbone, yielding a 512-dimensional spatiotemporal feature vector. The Optical Flow branch receives a 10-channel flow magnitude stack through a 2D ResNet-18, yielding a 512-dimensional motion feature vector. Because ImageNet-pretrained ResNet-18 expects 3-channel input, we replace its first convolutional layer with a 10-channel layer initialized by averaging the pretrained weights across channels and tiling; all subsequent layers retain pretrained weights. The two vectors are concatenated into a 1024-dimensional joint representation and passed through FC-512 $\rightarrow$ ReLU $\rightarrow$ Dropout(0.5) $\rightarrow$ FC-101 to produce class logits.

B. Optical Flow Computation

Dense optical flow is computed via OpenCV's Farneback implementation [4]. Although 16 input frames yield 15 consecutive frame-pairs, we uniformly sample 10 of the 15 pairs to obtain a fixed-size flow representation; this reduces Farneback computation by one-third with negligible loss in motion coverage. During training, the 10 flow magnitude maps are stacked into a tensor of shape [10, 224, 224]. At inference, frames are first downsampled to 112×112, yielding [10, 112, 112], reducing computation without significant loss in motion discriminability. Flow is computed asynchronously on a background thread during inference.

C. Real-Time Inference Strategy

To meet the $\geq$15 FPS target on a single NVIDIA RTX 5060 Ti, three optimizations are applied: (1) clip subsampling—every other frame is processed, reducing temporal resolution from 16 to 8 effective frames; (2) resolution reduction—input downsampled from 224×224 to 112×112; (3) FP16 inference—PyTorch AMP halves memory bandwidth on modern GPUs. Together these drop per-clip latency from $\sim$110 ms to 43.9 ms (22.8 FPS).

D. Training Details

Both branches are initialized from ImageNet-pretrained weights and fine-tuned end-to-end on UCF-101 with SGD (momentum 0.9, LR 0.001 decayed $\times 0.1$ at epochs 30 and 45, weight decay 5×10^{-4}, batch 32, 60 epochs). Augmentation includes random crop, horizontal flip, and color jitter. Training used an NVIDIA RTX 5060 Ti with PyTorch 2.1; total training time was approximately 14 hours.

IV. Dataset

We evaluate on UCF-101 [5], a standard benchmark containing 13,320 video clips spanning 101 action categories including sports, musical instruments, body motion, and human-object interactions. The dataset presents meaningful intra-class variation due to differences in viewpoint, background clutter, illumination, and camera motion. We follow the official three-fold cross-validation protocol and report mean accuracy across all three splits.

V. Experiments and Results

A. Main Results on UCF-101

Table I reports Top-1 and Top-5 accuracy for three model variants and our final fusion model. The FP32 fusion model achieves 87.4% Top-1, outperforming both the 3D CNN-only (82.1%) and Optical Flow-only (39.62%) baselines. Fairness caveat: the two single-branch baselines use 16-frame clips, whereas the fusion variants use 8-frame clips (inference-time subsampling). The fusion model's FPS advantage therefore partially reflects the halved temporal input rather than architectural efficiency alone. We do not report a 16-frame fusion speed figure as it falls below the real-time target and is outside the scope of this deployment-oriented evaluation; readers should interpret the FPS column as the practical real-time operating point rather than a controlled architectural comparison.

Model	Top-1 (%)	Top-5 (%)	Fr.	FPS
3D CNN Only (C3D-ResNet18)	82.1	95.3	16	31.2
Optical Flow + 2D CNN Only†	39.62	65.08	16	18.4
Fusion FP32 (Full)	87.4	97.1	8*	17.6
Fusion FP16 (Ours)	87.2	97.0	8*	22.8

Table 1: UCF-101 results (mean, 3 splits). *8-frame clip subsampling. †Split 1 only.

B. Inference Speed Analysis

Table II details per-clip latency on an RTX 5060 Ti (8-frame clips, 112×112, FP16). Optical flow computation is the dominant bottleneck in the sequential implementation. Summing the non-flow stages yields 37.2 ms; the measured total of 43.9 ms leaves a 6.7 ms gap attributable to thread synchronization, CUDA stream launch latency, and GPU memory transfer between the async flow thread and the inference pipeline—costs inherent to the pipelined design.

Pipeline Stage	ms	%
Frame capture & decode	5.2	9.3
Optical flow (Farneback)	18.7	33.5
RGB branch (3D ResNet-18)	14.3	25.6
Flow branch (2D ResNet-18)	10.1	18.1
Fusion FC layers	7.6	13.6
Total (sequential)	55.9	100
Total (async + FP16)	43.9	—

Table 2: Latency breakdown, Fusion model, RTX 5060 Ti.

C. Ablation Study

Table III presents ablations on Split 1. The Split 1 baseline is 88.1%, slightly above the 3-split mean of 87.4% due to favorable class distribution in this split. Disabling dropout causes a 1.9-point drop; removing ImageNet pretraining costs 4.6 points, confirming the value of transfer learning. Using 16-frame clips at inference recovers 1.2 points but reduces FPS below the real-time threshold.

Configuration	Top-1 (%)	Δ
Full Fusion (Split 1 baseline)	88.1	—
w/o Dropout	86.2	-1.9
w/o ImageNet Pretraining	83.5	-4.6
16-frame clips at inference	89.3	+1.2
FP32 only (no FP16)	88.1	0.0
No async flow (sequential)	88.1	0.0†

Table 3: Ablation study on Split 1. †Speed impact only.

D. Qualitative Analysis

The model performs well on actions with distinctive motion signatures such as archery, basketball dunk, and hammer throw, where the optical flow branch provides strong discriminative signal. The most common failure cases occur between visually and dynamically similar pairs: bowling vs. throwing, pull-ups vs. push-ups, and rock climbing indoor vs. rock climbing. These errors suggest occasional reliance on background context rather than action-specific motion, which could be mitigated with background suppression or attention-based localization.

VI. Discussion and Summary

The results confirm the central hypothesis: fusing 3D CNN spatiotemporal features with explicit optical flow yields meaningfully better accuracy than either branch alone. The 5.3-point improvement over the RGB-only baseline is consistent with two-stream literature, and demonstrates that Farneback optical flow—despite being a classical algorithm—provides complementary signal to learned 3D features.

The FP16 fusion model achieves 22.8 FPS, exceeding the 15 FPS real-time threshold. This is a practically meaningful result: the accuracy benefit of fusion does not come at the cost of real-time applicability. The primary remaining bottleneck is optical flow computation; future work could explore fast learned-flow networks (e.g., RAFT-Tiny) as a drop-in replacement.

A limitation is the exclusive evaluation on UCF-101, a relatively small benchmark. Performance on noisier, uncurated video (e.g., Kinetics-400) would require larger models and longer training schedules. Additionally, the system classifies pre-trimmed clips and cannot detect or localize actions in untrimmed video.

In summary, we presented a real-time two-branch action recognition system combining a lightweight 3D CNN with an optical flow branch. With FP16 inference and 8-frame subsampling, the system achieves 87.2% Top-1 accuracy at 22.8 FPS on UCF-101 with a single consumer GPU. Ablation experiments confirm the value of both branches, transfer learning, and temporal sampling strategy. Future directions include replacing Farneback flow with a learned fast-flow network and extending evaluation to large-scale benchmarks such as Kinetics-400.

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